

Sally wants to know how much more she is expected to make if she studies one more year. She knows that the effect of education may vary with years of education, so she decides to run an econometric model with the following nonlinear regression function:

$$ahe_i = \beta_0 + \beta_1 yrseduc_i + \beta_2 yrseduc_i^2 + \beta_3 female_i + \beta_4 age_i + u_i,$$

where

ahe_i = Average Hourly Earnings in U.S. dollars;

$yrseduc_i$ = Years of Education;

$female_i$ = Binary Variable (1 if Female; 0 if Male);

age_i = Age (in Years);

and gets the output:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	3.28361	0.80212	4.09	0.0000425	***
yrseduc	-0.55862	0.12132	-4.60	0.0000041	***
I(yrseduc^2)	0.10564	0.00465	22.71	< 2e-16	***
female	-5.63909	0.09162	-61.55	< 2e-16	***
age	0.18954	0.00401	47.22	< 2e-16	***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Based on this model and the estimates, please answer:

1. Sally has 18 years of education. How much different would her expected average hourly earnings be (i.e. find the expected difference Δahe_i due to $\Delta yrseduc_i = 1$), had she studied 19 years by now (i.e. age does not change)?
2. What about her son, William, who has studied 12 years already? Find the expected difference Δahe_i due to $\Delta yrseduc_i = 1$ in his case, while also holding other variables fixed.
3. Please discuss how one can obtain the 95% confidence interval for the expected difference, Δahe_i , you obtained for Sally. You do not need to provide (in fact, you cannot based on this output only) a numerical answer.

SOLUTION:

$$1) \hat{ahe} | yrseduc=19 - \hat{ahe} | yrseduc=18 =$$

$$(\cancel{\beta_0} + \beta_1 \cdot 19 + \beta_2 \cdot 19^2 + \cancel{\beta_3(1)} + \cancel{\beta_4(\text{age}_{sally})}) -$$

$$(\cancel{\beta_0} + \beta_1 \cdot 18 + \beta_2 \cdot 18^2 + \cancel{\beta_3(1)} + \cancel{\beta_4(\text{age}_{sally})}) =$$

$$(\beta_1 \cdot 19 + \beta_2 \cdot 19^2) - (\beta_1 \cdot 18 + \beta_2 \cdot 18^2) =$$

$$(-0.55862 \cdot 19 + 0.10564 \cdot 19^2) -$$

$$(-0.55862 \cdot 18 + 0.10564 \cdot 18^2) =$$

$$(-10.61378 + 38.13604) -$$

$$(-10.05516 + 34.22736) =$$

$$27.52226 - 24.1722 = 3.35006$$

$$\rightarrow \text{so, } \hat{ahe} \text{ from 18 to 19 yrseduc} = \boxed{\$ 3.35006 / \text{hour}}$$

$$2) \hat{ahe} | yrseduc=13 - \hat{ahe} | yrseduc=12 =$$

$$(\cancel{\beta_0} + \beta_1 \cdot 13 + \beta_2 \cdot 13^2 + \cancel{\beta_3(0)} + \cancel{\beta_4(\text{age}_{william})}) -$$

$$(\cancel{\beta_0} + \beta_1 \cdot 12 + \beta_2 \cdot 12^2 + \cancel{\beta_3(0)} + \cancel{\beta_4(\text{age}_{william})}) =$$

$$(\beta_1 \cdot 13 + \beta_2 \cdot 13^2) - (\beta_1 \cdot 12 + \beta_2 \cdot 12^2) =$$

$$(-7.26206 + 17.85316) - (-6.70344 + 15.21216) =$$

$$10.5911 - 8.50872 = 2.08238$$

$$\rightarrow \text{so, } \hat{ahe} \text{ from 12 to 13 yrseduc} = \boxed{\$ 2.08238 / \text{hour}}$$

3) To find the 95% confidence interval for the expected difference, $\Delta a \hat{e}_i$ for Sally, you would first find d , which in the case of Sally $\Rightarrow d = [0 \ (19-18) \ (19^2-18^2) \ 0 \ 0] = [0 \ 1 \ 37 \ 0 \ 0]$ and then plug d^T into the following formula: to find the standard error:

$$d^T (d^T \hat{\beta}) \pm z_{\alpha/2} \sqrt{d^T \hat{\Sigma}_{\hat{\beta}} d} \quad \text{with } (1-\alpha) \cdot 100\% \text{ confidence}$$

$\underbrace{\hat{\Sigma}_{\hat{\beta}}}_{\text{variance matrix}} \rightarrow \text{covariance matrix}$